

ADVANCED MOTORSPORT SIMULATION

Simulation and Analysis of the Effect of Rear-Wing Angle on F1 Car Performance

Aerodynamics · Vehicle Dynamics · Lap-Time Optimization · Active Aero

Project	Advanced F1 Rear-Wing Angle Study
Model version	1.0.0
Run ID	6bf25b9bebec26e
Generated	2026-10-04T14:29:58.859901+00:00
Classification	Representative educational model
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Academic engineering report. Numerical values are traceable to the manifest and must be interpreted within the stated assumptions and limitations.

Document Control

Title	Simulation and Analysis of the Effect of Rear-Wing Angle on F1 Car Performance
Purpose	Reproducible advanced educational engineering study
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Random seed	3106
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Evidence bundle	results/manifest.json and SHA-256 artifact register
MATLAB status	R2022b+-compatible source; not runtime-tested in authoring environment

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Abstract

Rear-wing incidence is one of the clearest examples of a motorsport design trade-off: increasing angle generally raises downforce and cornering potential, while also raising drag and reducing straight-line performance. This project quantifies that trade-off for a representative Formula 1-style car using a reproducible quasi-steady vehicle model, three synthetic circuit archetypes, fixed and active aerodynamic strategies, bounded optimization, and uncertainty analysis. The complete workflow is implemented in Python and mirrored in MATLAB R2022b+-compatible source, with an optional programmatic Simulink demonstration.

The nominal fixed-wing optima are 17.996 degrees for the low-downforce circuit, 4.997 degrees for the balanced circuit, and 18.000 degrees for the high-downforce circuit. Their corresponding predicted lap times are 68.150 s, 58.812 s, and 64.528 s. The difference between circuits is more important than the label attached to a setup: optimum incidence emerges from the distribution of straights, corner radii, braking zones, tyre limits, and power-limited speed. The ideal-active model can improve selected cases, whereas actuator delay, finite slew rate, and simple switching logic may remove that benefit. Accordingly, the study's main conclusion is not that one angle is universally best, but that wing angle must be optimized as part of a coupled car-track-control system.

All numerical values are traceable to run ID 6bf25b9bebc26e. This is a representative educational model; it does not use proprietary team data, measured wind-tunnel maps, full-car CFD, or real circuit telemetry. Its purpose is comparative engineering analysis, transparent derivation, and a defensible platform for future physical validation.

Table 1. Study scope and outputs.

Item	Definition
Independent variable	Rear-wing angle, 0–30 deg
Circuit cases	Low-downforce, balanced, high-downforce synthetic closed tracks
Strategies	Fixed, open-wing, ideal-active, actuator-limited active
Primary response	Lap time with straight-line, braking, cornering, energy diagnostics
Uncertainty	1,000 bounded samples per circuit; seed 3106
Deliverables	CSV, XLSX, MAT, PNG, SVG, DOCX, PDF, PPTX, MATLAB and Simulink builder

1. Introduction and Research Question

1.1 Motivation

A racing car's aerodynamic package converts vehicle speed into forces that can enlarge the tyre force envelope. Downforce is valuable because it increases normal load without the same inertial penalty as adding mass. Drag is costly because it consumes tractive power and reduces acceleration and maximum speed. A rear wing influences both terms and also shifts aerodynamic balance toward the rear axle. The engineering problem is therefore coupled: the setting that is attractive in a slow corner may be unattractive on a long straight, and an aggressive setting can reach diminishing returns or stall.

Modern regulations reinforce the relevance of this question. FIA technical rules tightly define aerodynamic component volumes, construction, flexibility, and operation [1]. The 2026 framework also introduced coordinated movable front and rear aerodynamic states, described publicly as a higher-downforce cornering mode and a lower-drag straight-line mode [2]. This project does not claim regulatory fidelity to any season; the rules are used to establish context for why configurable aerodynamics and actuation limits matter.

1.2 Research question

The principal question is: how does rear-wing angle between 0 and 30 degrees affect aerodynamic coefficients, axle loads, straight-line acceleration, braking, cornering speed, energy demand, and total lap time for different circuit characteristics? Supporting questions ask whether an ideal active wing is materially superior to the best fixed setting, whether a rate- and delay-limited actuator preserves that gain, and which uncertain parameters most affect the selected optimum.

1.3 Objectives and success criteria

The work develops a transparent coefficient model, couples it to tyre and powertrain limits, solves closed laps through spatial forward/backward passes, and exports every plotted result as machine-readable data. Success requires physically interpretable trends, numerical convergence, deterministic regeneration, cross-format consistency, explicit limitations, and a MATLAB/Simulink handoff. Absolute agreement with a current Formula 1 car is not a success criterion because the required proprietary data are unavailable.

Table 2. Principal nomenclature.

Symbol	Meaning	Unit
A	Full-car aerodynamic reference area	m ²
AR	Rear-wing aspect ratio	—
C _L	Positive downforce coefficient magnitude	—
C _D	Drag coefficient	—
e	Span-efficiency factor	—
F _z	Normal load	N
μ	Tyre friction coefficient	—
q	Dynamic pressure	Pa
ρ	Air density	kg/m ³
v	Vehicle speed	m/s

2. Literature Review

2.1 Race-car aerodynamic trade-offs

The conventional force definitions express lift and drag as coefficients multiplied by dynamic pressure and reference area [3]. For a racing car, the useful vertical force is downward; this report therefore uses a positive downforce coefficient magnitude. Finite wings also generate induced drag. A standard parabolic relation scales induced drag with the square of lift coefficient and inversely with aspect ratio and span efficiency [4]. These relations do not replace detailed aerodynamics, but they are appropriate building blocks for a transparent conceptual model.

Race-car literature emphasizes that aerodynamics must be evaluated through vehicle performance rather than coefficient magnitude alone [7]. A configuration with maximum downforce need not minimize lap time because its drag penalty acts over every high-speed metre. Conversely, maximum lift-to-drag ratio need not be the lap-time optimum because braking and cornering depend on absolute load as well as efficiency. This distinction motivates the integrated simulations used here.

2.2 Lap-time simulation

Published work has connected CFD-derived wing coefficients to transient or quasi-steady lap models to estimate the effect of adding wings [5]. Established lap-time methods range from steady-state envelopes through quasi-static spatial solvers to full transient optimal-control formulations [10,11]. The chosen forward/backward method occupies the middle ground: it retains local track curvature, speed-dependent forces, gearing, acceleration, and braking, while remaining auditable and fast enough for angle sweeps and uncertainty studies.

2.3 Movable wings and actuator realism

Research on Formula Student drag-reduction systems demonstrates that movable elements can reduce drag, but also shows that actuator packaging itself can disturb lift and drag [6]. This is a useful caution: an aerodynamic state change cannot be assessed independently from the mechanism, command delay, transition time, and safety logic. The present study therefore reports both an ideal-active upper bound and an actuator-limited case. The latter is intentionally simple, but it prevents the common mistake of assigning instantaneous, cost-free switching.

2.4 Tyre and vehicle coupling

Tyre force capacity varies with normal load and combined-slip demand [8,9]. Increasing downforce raises available tyre force, but tyre load sensitivity means the gain is sub-linear. Aerodynamic balance also matters because the limiting axle can saturate before the sum of nominal front and rear capacities. This project evaluates front and rear normal loads separately, then applies a load-sensitive friction ellipse. It thereby captures the first-order reason why rear-wing angle can change stability margin and performance even at identical total downforce.

2.5 Identified gap and contribution

Many educational demonstrations stop at plots of lift and drag versus angle. This project extends the chain from wing setting to coefficient, force, axle load, tyre envelope, wheel force, speed trace, lap time, energy, optimization, uncertainty, and reportable evidence. It contributes an open, reproducible multi-language workflow rather than a claim of high-fidelity Formula 1 prediction.

3. Aerodynamic Theory

3.1 Dynamic pressure and force coefficients

For air density ρ and vehicle speed v , dynamic pressure q is:

$$q = \frac{1}{2} \rho v^2$$

Equation (1)

Downforce and drag follow from the full-car coefficients and reference area A :

$$F_{\text{down}} = q A C_L$$

Equation (2)

$$F_{\text{drag}} = q A C_D$$

Equation (3)

These expressions explain the strong speed dependence visible in the force sweeps. Doubling speed multiplies the aerodynamic force by four if coefficients remain fixed. The effect of wing angle therefore grows rapidly in the fastest corners and braking zones.

3.2 Finite-wing lift slope

The rear wing is represented by a corrected finite-wing lift-curve slope. With two-dimensional slope a_0 , aspect ratio AR , and span efficiency e :

$$a = a_0 / (1 + a_0 / (\pi e AR))$$

Equation (4)

Aspect ratio is obtained from span b and wing planform area S_w :

$$AR = b^2 / S_w$$

Equation (5)

The pre-stall wing coefficient is linear in effective incidence relative to zero lift. Beyond the configured stall angle, a smooth exponential decay avoids a non-physical discontinuity. An additional smooth drag rise represents separated-flow losses. These post-stall terms are empirical assumptions and should be replaced with measured polar data when available.

3.3 Drag polar

Before stall, wing drag is modeled as profile drag plus induced drag:

$$C_{D,w} = C_{D0,w} + C_{L,w}^2 / (\pi e AR)$$

Equation (6)

The wing coefficients are area-scaled and added to fixed full-car baseline coefficients. The total model is deliberately modular: a CFD or wind-tunnel lookup table can replace the analytical polar without rewriting the vehicle solver.

RESULT PLATE 01

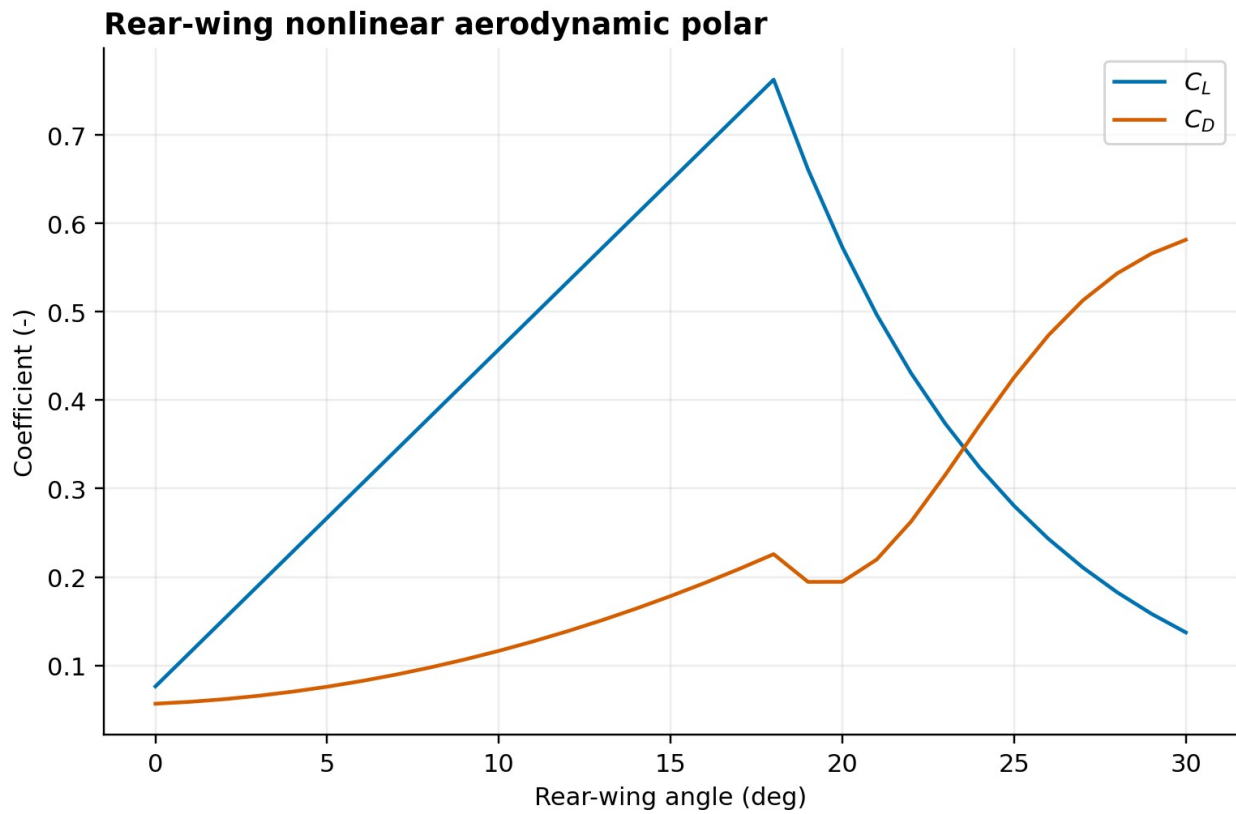


Figure 1. Wing lift and drag coefficients, including the modeled stall transition.

Source data: data/aero_polar.csv · Units: dimensionless · Run: 6bf25b9bebc26e

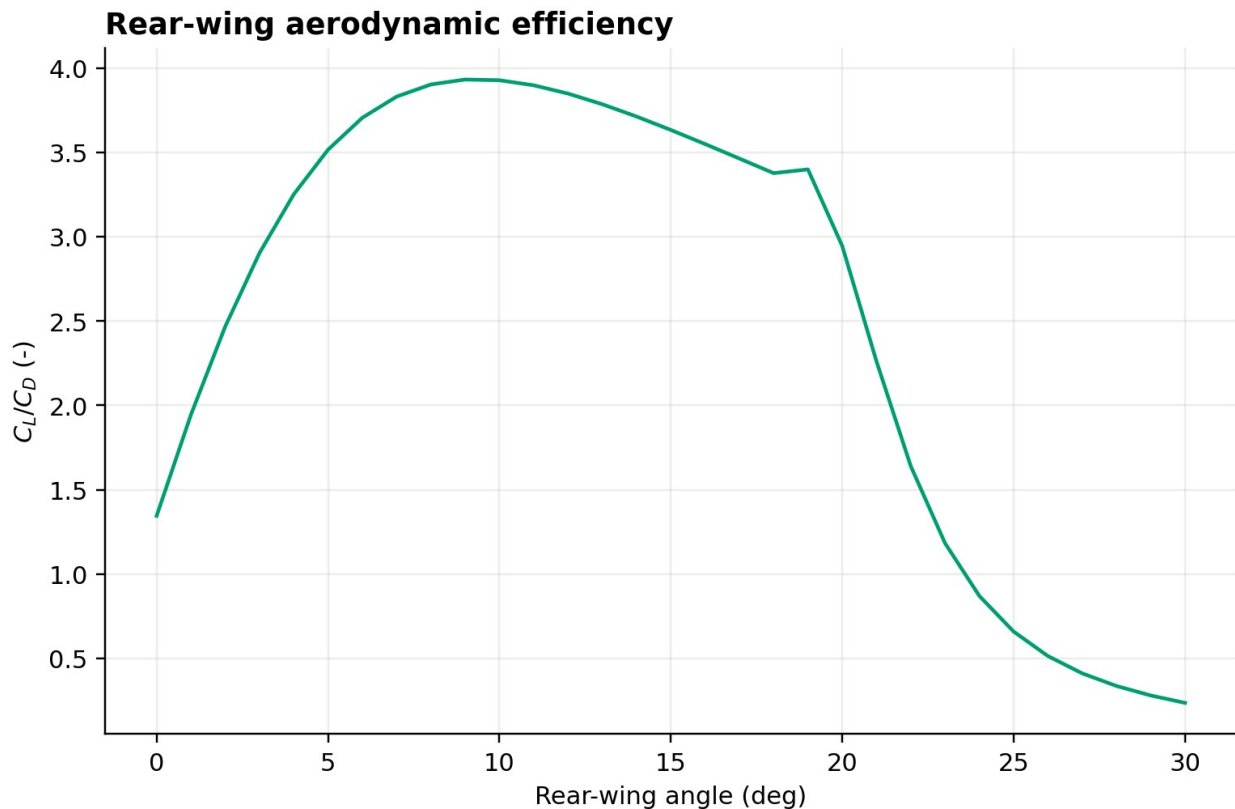
RESULT PLATE 02

Figure 2. Aerodynamic efficiency peaks before maximum downforce and falls through stall.

Source data: data/aero_polar.csv · Units: dimensionless · Run: 6bf25b9bebcec26e

3.4 Aerodynamic balance

Baseline and wing forces act at declared longitudinal application points. Static moments about the axles distribute aerodynamic load between front and rear. Raising rear-wing incidence increases total downforce and typically shifts the aerodynamic load fraction rearward. This can increase rear traction but make the front axle the cornering constraint. The model therefore retains axle-resolved loads rather than using only total downforce.

RESULT PLATE 03

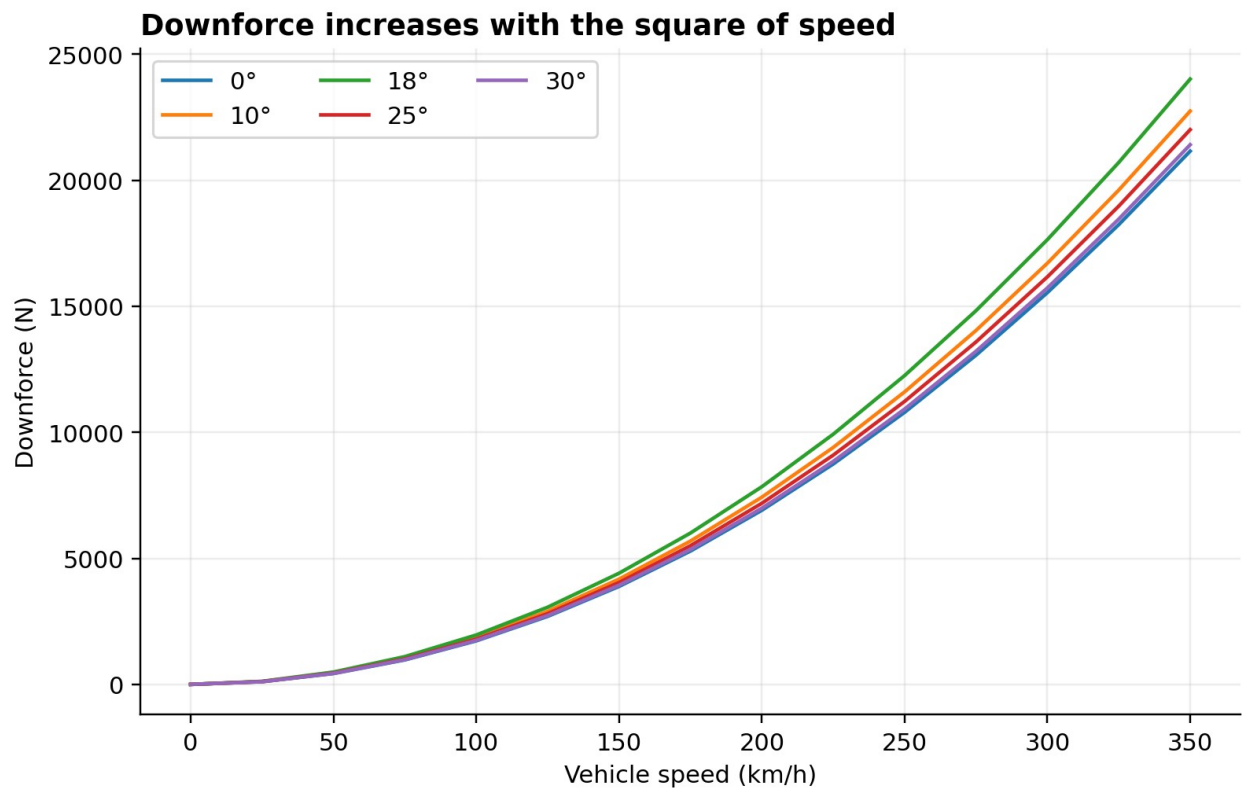


Figure 3. Downforce increases with the square of speed for selected rear-wing angles.

Source data: data/force_speed.csv · Units: N · Run: 6bf25b9bebc26e

RESULT PLATE 04

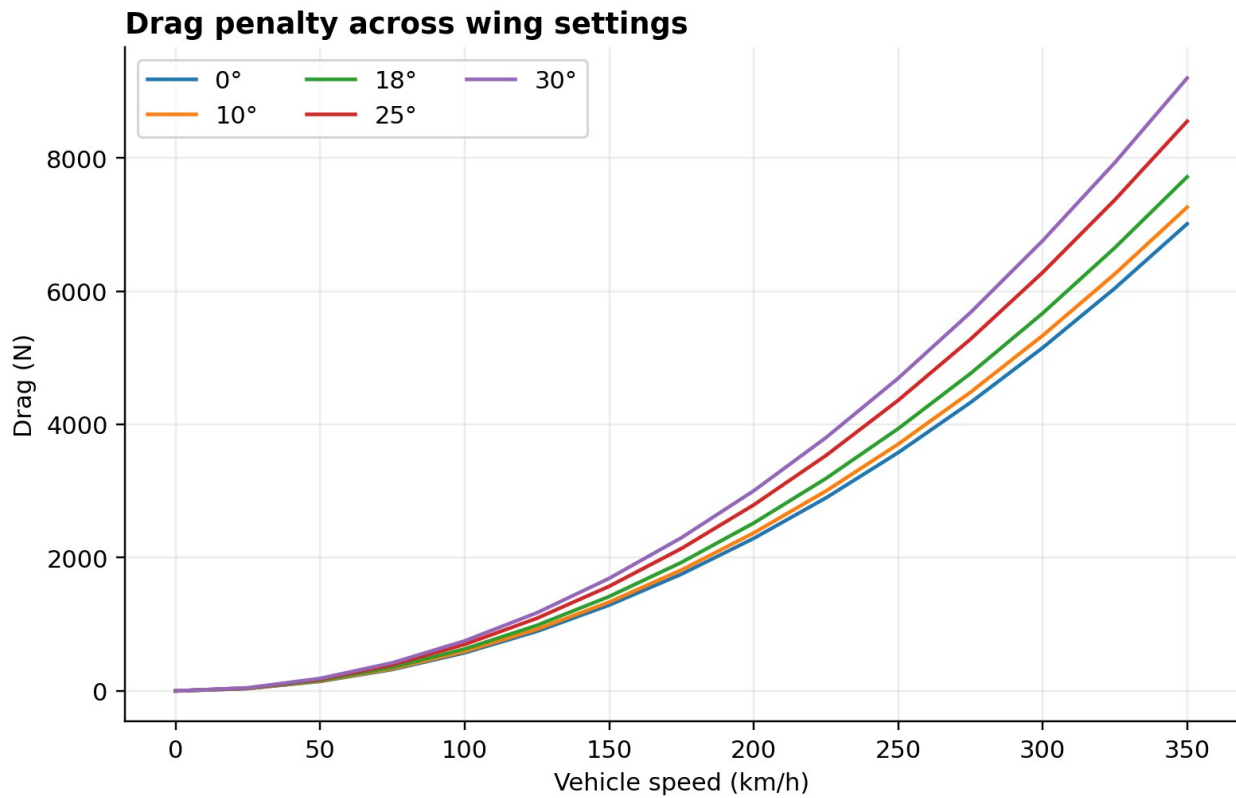


Figure 4. Drag penalty across wing settings for selected rear-wing angles.

Source data: data/force_speed.csv · Units: N · Run: 6bf25b9bebec26e

RESULT PLATE 05

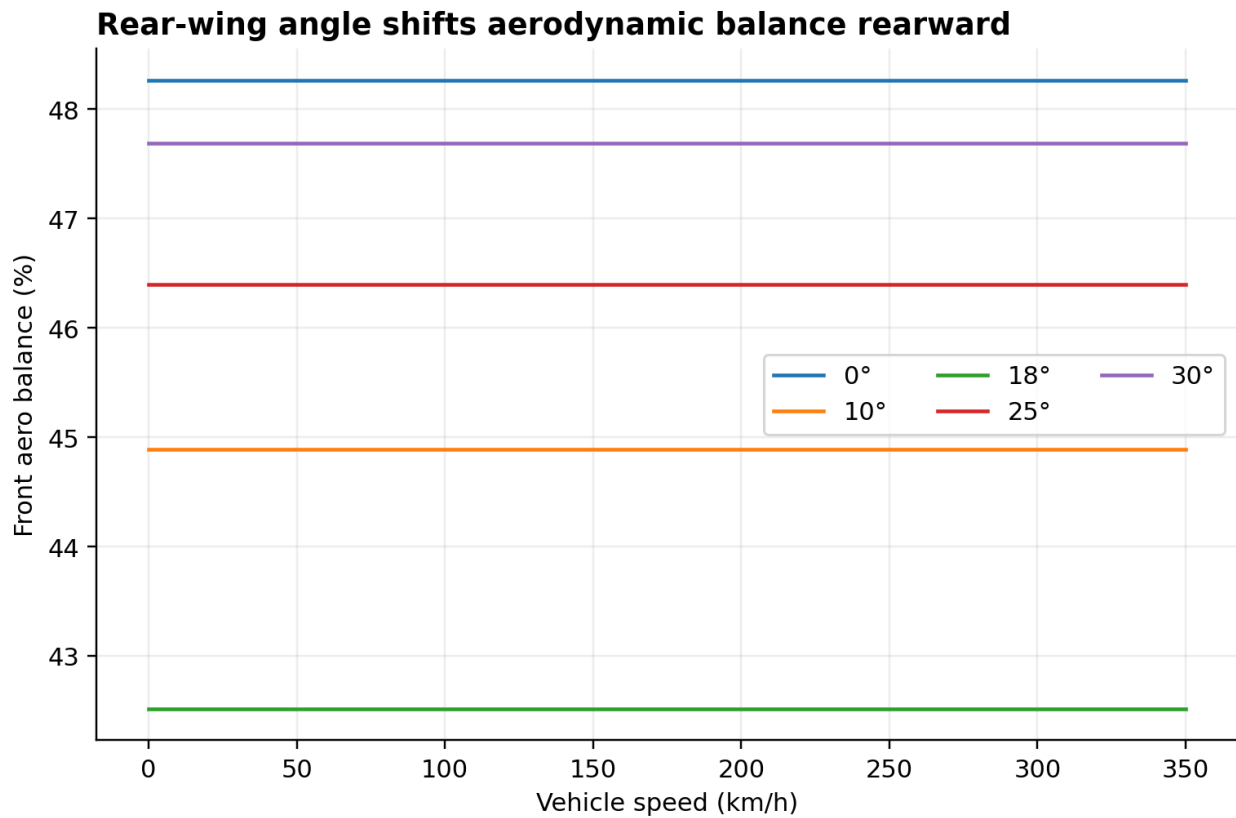


Figure 5. Front share of total aerodynamic load for selected rear-wing angles.

Source data: data/aero_balance.csv · Units: % · Run: 6bf25b9bebec26e

4. Vehicle and Tyre Dynamics

4.1 Normal load and load transfer

Static axle loads are determined by mass distribution. Longitudinal acceleration introduces a load-transfer term based on mass m , acceleration a_x , centre-of-gravity height h , and wheelbase L :

$$\Delta F_z = m a_x h / L$$

Equation (7)

Speed-dependent aerodynamic loads are then added at each axle. Grade changes the gravity component along the path and slightly changes the normal component. Pitch-heave aerodynamic coupling is omitted at nominal conditions, which should be remembered when interpreting very high-speed braking.

4.2 Load-sensitive tyre capacity

The reference friction coefficient is scaled using an exponent n :

$$\mu(F_z) = \mu_{\text{ref}} (F_z / F_{z,\text{ref}})^{-n}$$

Equation (8)

Because n is positive, friction coefficient falls as load rises. Total force still increases with load, but less than proportionally. This is why downforce remains valuable while showing diminishing returns. Combined longitudinal and lateral demands use a generalized friction ellipse:

$$(F_x / F_{x,\text{max}})^p + (F_y / F_{y,\text{max}})^p \leq 1$$

Equation (9)

Separate front and rear checks prevent the model from borrowing unused capacity from one axle after the other axle has saturated.

4.3 Powertrain and resistance

Engine torque is linearly interpolated across a representative speed curve. Each gear maps vehicle speed to engine speed; the solver selects the admissible gear giving maximum wheel force. Wheel force is limited by rear-axle traction. Longitudinal resistance is:

$$F_{\text{res}} = F_{\text{drag}} + C_{\text{rr}} (mg \cos(\theta) + F_{\text{down}}) + mg \sin(\theta)$$

Equation (10)

Top speed occurs where available wheel force equals total resistance. Acceleration integrates the net force; braking combines the friction envelope and aerodynamic/rolling resistance. Energy reported here is positive tractive wheel work, not fuel energy.

RESULT PLATE 06

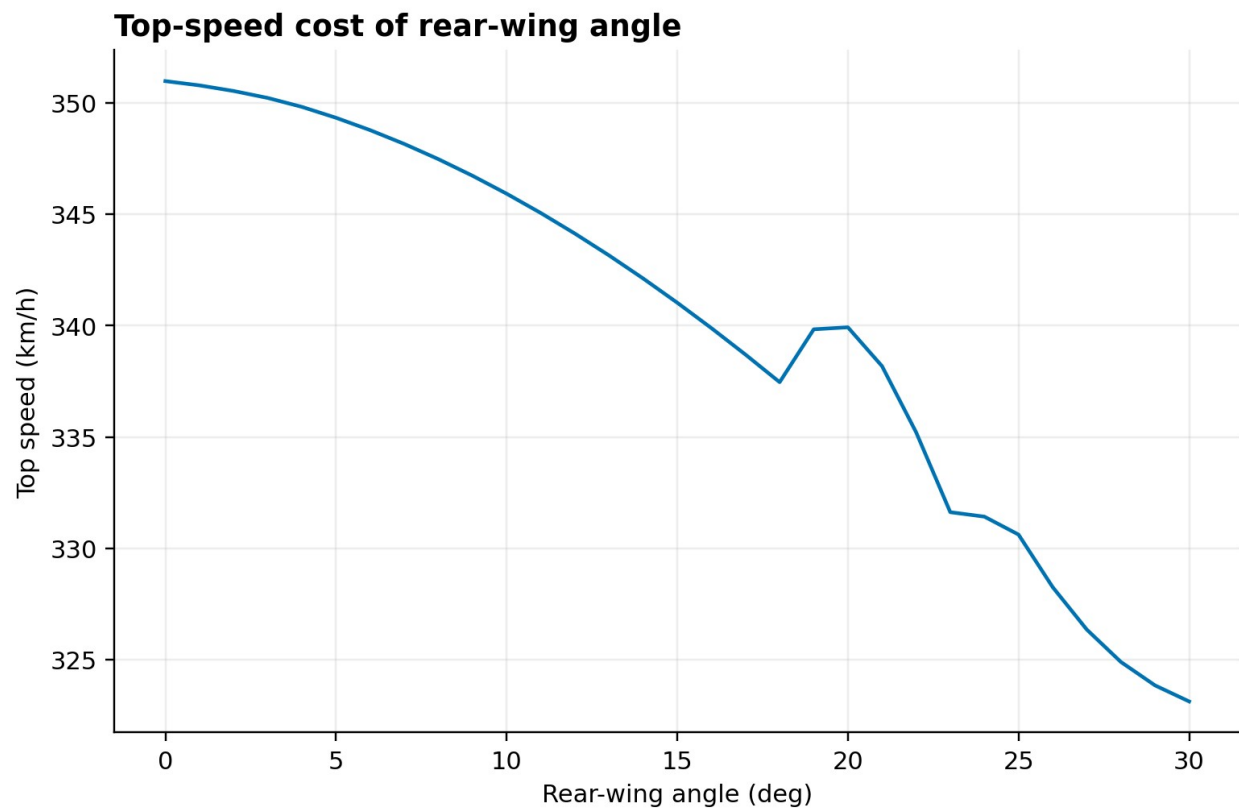


Figure 6. Power-limited top speed decreases as wing drag rises.

Source data: data/straight_line.csv · Units: km/h · Run: 6bf25b9bebec26e

RESULT PLATE 07

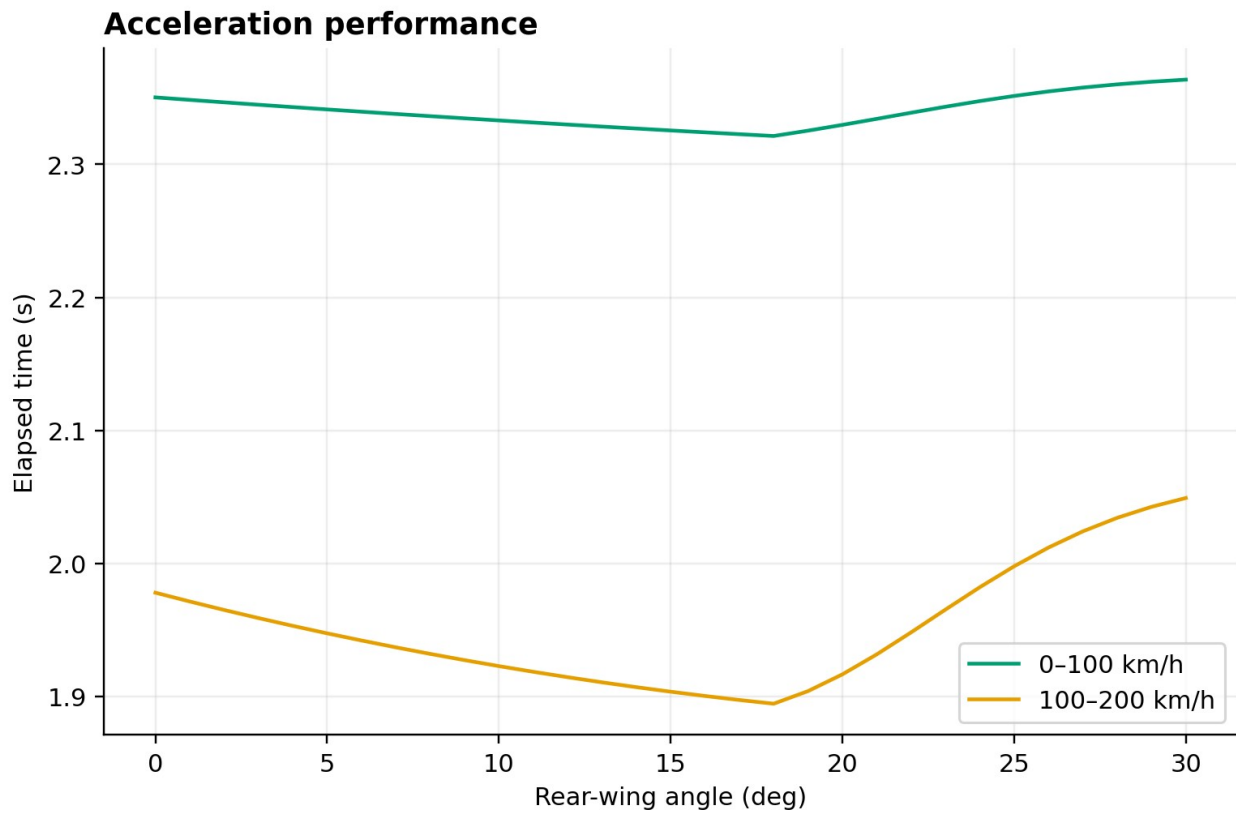


Figure 7. Acceleration time includes gearing, traction, rolling resistance, and aerodynamic drag.

Source data: data/straight_line.csv · Units: s · Run: 6bf25b9bebcec26e

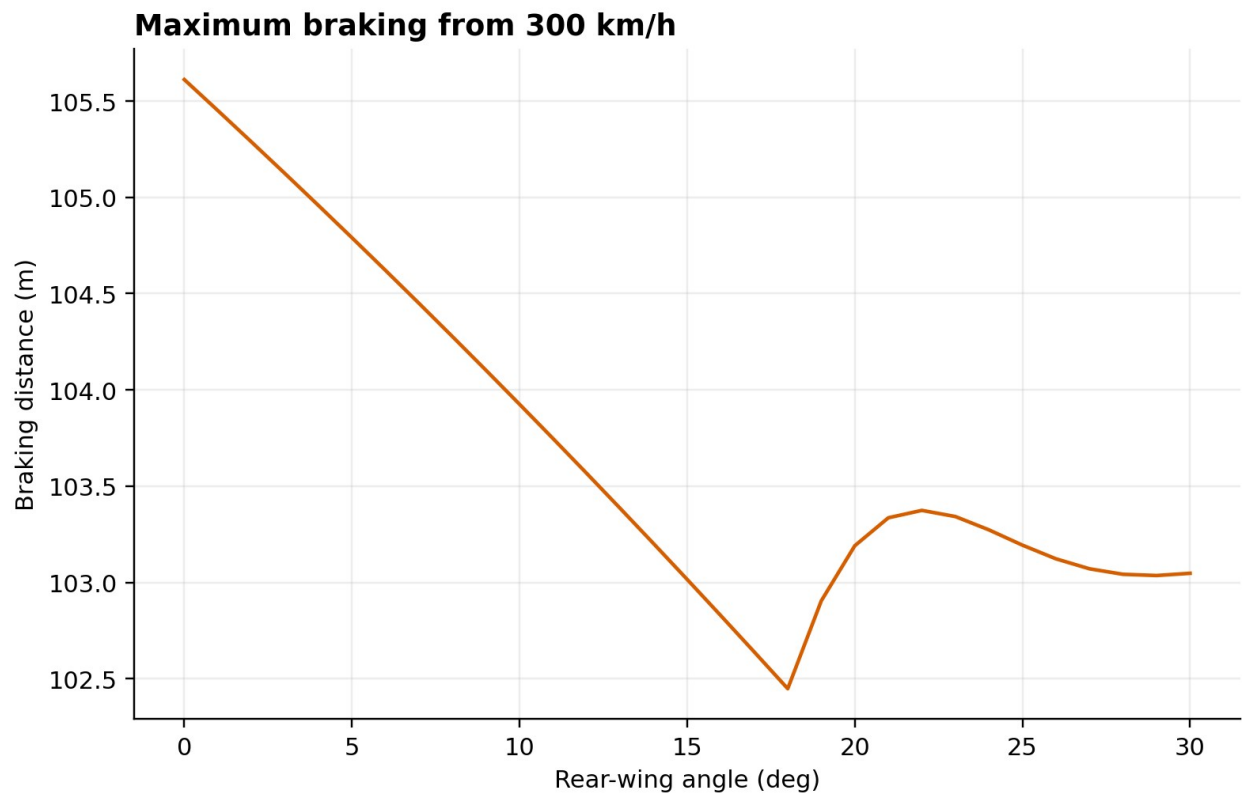
RESULT PLATE 08

Figure 8. Integrated 300-0 km/h braking distance with load-sensitive tyres and aero.

Source data: data/straight_line.csv · Units: m · Run: 6bf25b9bebc26e

RESULT PLATE 09

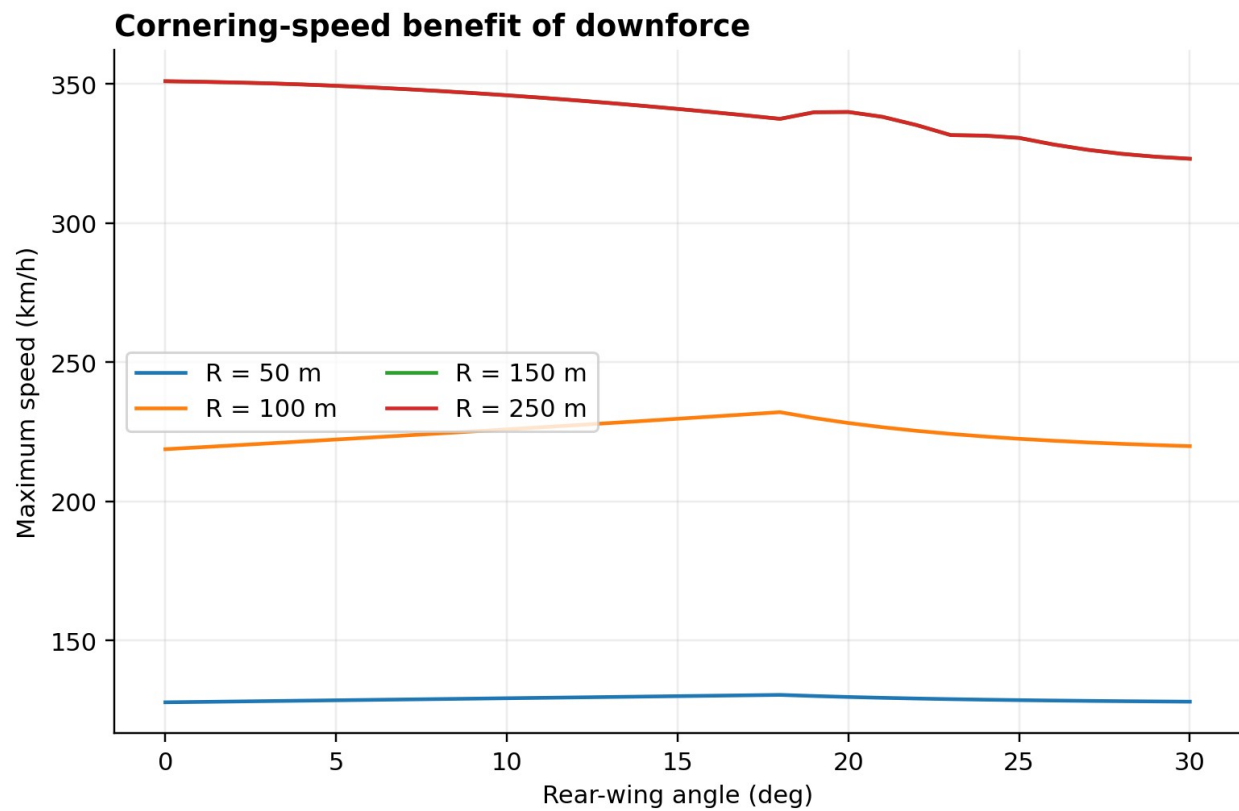


Figure 9. Axle-constrained maximum cornering speed for representative radii.

Source data: data/cornering.csv · Units: km/h · Run: 6bf25b9bebcec26e

5. Model Architecture and Implementation

5.1 Data flow

The configuration is stored in versioned JSON, validated into typed Python data structures, and consumed by independent aerodynamic, dynamics, lap, strategy, analysis, and export modules. Track files use segments with length, entry and exit curvature, grade, active-aero eligibility, and optional speed limit. Each segment is discretized at no more than 5 m. The output manifest binds inputs, source tables, plots, headline metrics, random seed, and SHA-256 hashes to a deterministic run identifier.

5.2 Spatial lap solver

For each spatial cell, a lateral-speed limit is found from axle tyre capacity. A forward pass then limits the speed reachable under power and traction; a backward pass limits the speed from which the next cell can be reached under braking. Passes repeat around the closed track until the maximum speed change falls below the tolerance. Segment time is computed using the trapezoidal speed relation:

$$\Delta t_i = 2 \Delta s_i / (v_i + v_{i+1})$$

Equation (11)

Lap time is the sum over all cells:

$$t_{\text{lap}} = \sum_i \Delta t_i$$

Equation (12)

This method is deterministic, interpretable, and efficient for many setup evaluations. It does not solve for the racing line or steering trajectory.

5.3 Circuit archetypes

Three synthetic closed circuits isolate contrasting demands. The low-downforce case emphasizes long high-speed sections, the balanced case mixes straights and medium-speed corners, and the high-downforce case increases corner density and curvature. Names describe intended character rather than guaranteeing a particular optimum; the actual optimum remains a result of the full speed/force distribution.

Table 3. Synthetic circuit definitions.

Track	Length (m)	Segments	Max curvature (1/m)	Aero-eligible segments
low downforce	5,820.0	9	0.012	5
balanced	5,400.0	10	0.010	4
high downforce	4,720.0	12	0.014	4

5.4 Fixed and active strategies

Fixed-angle analysis evaluates every integer angle from 0 to 30 degrees and refines the best region with bounded interpolation. Open-wing is a prescribed low-drag reference. Ideal-active switches without actuation cost and is interpreted only as an upper bound. Actuator-limited active aero applies delay, rate limit, and eligibility logic, then solves the resulting schedule. These strategies are compared using time, maximum speed, tractive energy, and full-throttle fraction.

6. Verification and Reproducibility

6.1 Verification hierarchy

Verification is layered. Unit tests check configuration validation, aerodynamic trends, stall behavior, force balances, load sensitivity, gearing, track closure, solver convergence, strategy constraints, uncertainty determinism, and exports. Analytical checks confirm the v-squared force relationship and basic limits. Regression fixtures lock representative aerodynamic and lap outputs. Cross-artifact checks compare the CSV, XLSX, MAT, manifest, report, and presentation values.

6.2 Numerical controls

The spatial step is 5 m, solver speed tolerance is 0.001 m/s, and the maximum iteration count is 200. Input ranges are validated before analysis. Optimization is bounded to the physical study domain. The Monte Carlo seed is fixed at 3106, so the same inputs produce identical samples and headline statistics. A deterministic run ID is derived from input hashes instead of time.

6.3 MATLAB parity and Simulink scope

MATLAB package functions load the same JSON configuration and tracks and reproduce the aerodynamic, tyre, wheel-force, lap, sweep, and export interfaces. `run_parity_tests.m` reads Python-generated reference cases covering all circuits and pre-/post-stall angles. Because MATLAB is unavailable in the authoring environment, this source is statically checked but not runtime-tested. Final parity should be executed in MATLAB R2022b+ [12]. `build_simulink_demo.m` programmatically creates a focused model of wing-angle command, rate limiting, aerodynamic forces, vehicle acceleration, logging, and scope output when Simulink is licensed.

Table 4. Verification evidence summary.

Check	Count	Evidence
Input provenance	5	SHA-256 registered
Output artifacts	51	SHA-256 registered
Numerical tables	16	CSV + XLSX mapping
Figures	16	PNG + SVG + source CSV
Randomness	1	Deterministic seed 3106
MATLAB parity	12	Static contract + reference fixtures

7. Results

7.1 Aerodynamic response

The analytical polar increases downforce with angle before the 18-degree stall threshold. Drag rises throughout and grows more sharply as separation is introduced. Aerodynamic efficiency peaks earlier than maximum downforce, illustrating why coefficient efficiency and lap-time optimum are different objectives. Force-speed curves confirm that setup differences become most consequential at high speed.

7.2 Straight-line and cornering effects

Higher wing angle reduces the power-limited top speed and generally lengthens high-speed acceleration. It can shorten braking distance because higher downforce enlarges tyre capacity while drag adds direct deceleration. Maximum corner speed rises where aerodynamic load is valuable, but axle balance and tyre load sensitivity prevent unlimited gains. These competing trends create the lap-time minima.

7.3 Fixed-wing optimization

The optimized fixed settings are summarized below. The low-downforce circuit selects 17.996 degrees and 68.150 s; the balanced circuit selects 4.997 degrees and 58.812 s; and the high-downforce circuit selects 18.000 degrees and 64.528 s. The balanced optimum is broad: its reported neighbor sensitivity is small, so a nearby setting can be operationally preferable if it improves stability, cooling, or robustness. The two 18-degree results lie at the modeled pre-stall peak; angles beyond this region are penalized by both lift decay and drag rise.

Table 5. Optimized fixed-wing results.

Track	Angle (deg)	Lap time (s)	Neighbour sensitivity (s/deg)	Method
low downforce	17.996	68.150	0.042	integer sweep + PCHIP bounded refinement
balanced	4.997	58.812	0.003	integer sweep + PCHIP bounded refinement
high downforce	18.000	64.528	0.087	integer sweep + PCHIP bounded refinement

RESULT PLATE 10

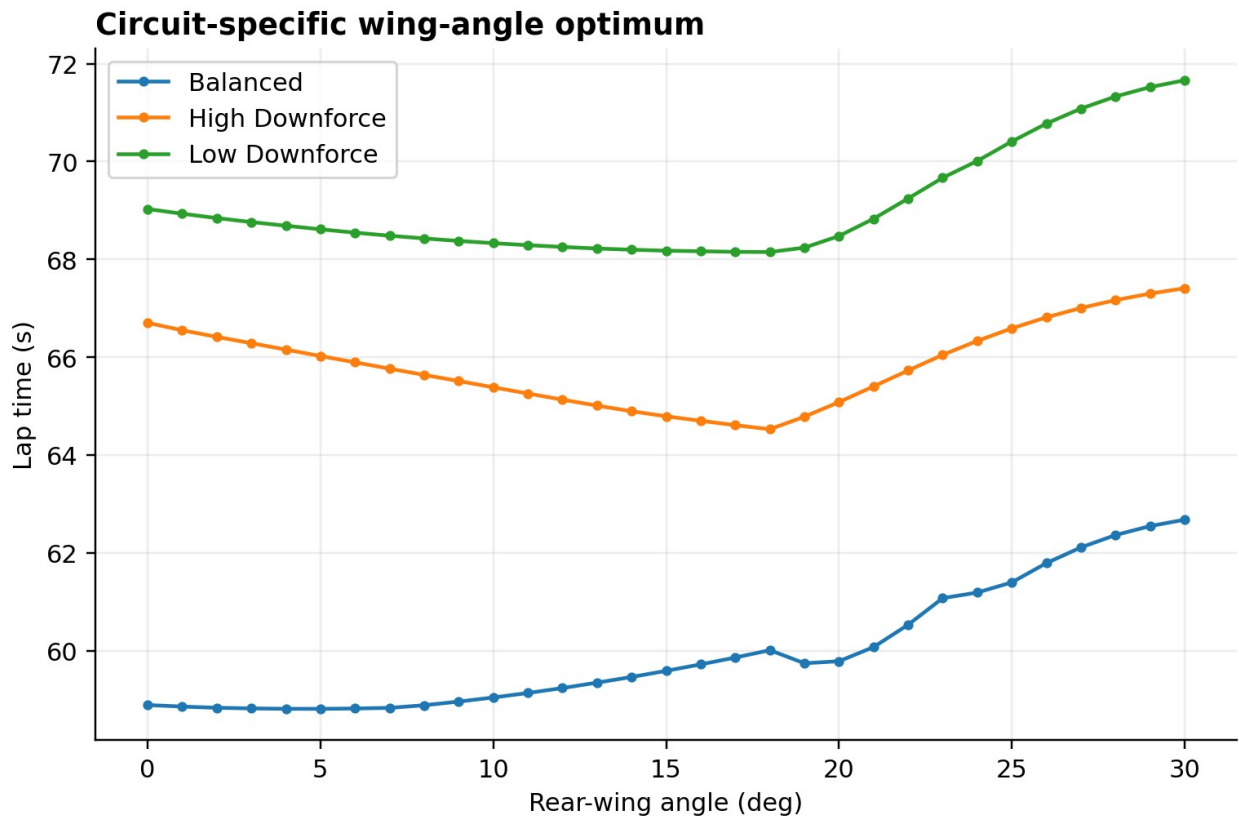


Figure 10. Fixed-wing lap-time sweep for low-, balanced-, and high-downforce circuits.

Source data: data/lap_sweeps.csv · Units: s · Run: 6bf25b9bebccec26e

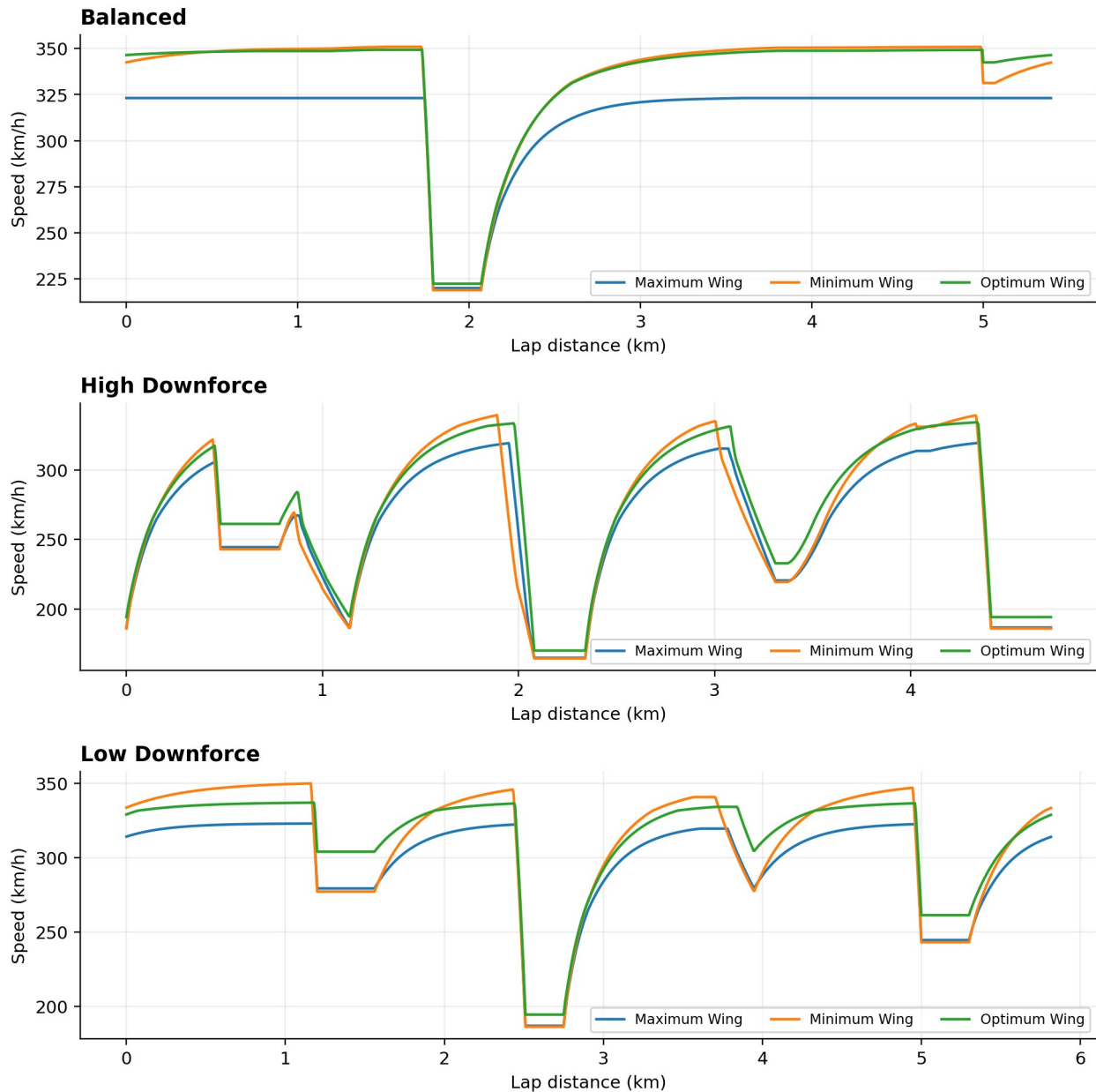
RESULT PLATE 11

Figure 11. Minimum-, optimum-, and maximum-wing speed traces on each circuit archetype.

Source data: data/speed_traces.csv · Units: km/h · Run: 6bf25b9bebec26e

7.4 Strategy comparison

The ideal-active strategy is best on the low- and high-downforce archetypes in this model, demonstrating the theoretical value of changing the drag/downforce compromise along the lap. Yet the actuator-limited strategy records 68.599 s, 59.353 s, and 64.615 s—slower than the optimum fixed setting on all three circuits. This is a meaningful negative result. Delay, slew, and the intentionally simple schedule cause time to be lost during transitions and can place the wing in a poor intermediate state. Active hardware alone does not guarantee lap-time benefit; the command policy must be co-designed and validated.

Table 6. Aerodynamic strategy comparison.

Track	Mode	Open (deg)	Closed (deg)	Lap (s)	Vmax (km/h)	Wheel work (MJ)
low downforce	fixed-optimum	17.996	17.996	68.150	336.87	37.130
low downforce	open-wing	2.000	20.000	67.952	340.11	35.275
low downforce	active-ideal	0.000	18.000	67.606	337.64	34.883
low downforce	active-limited	0.000	18.000	68.599	349.72	38.725
balanced	fixed-optimum	4.997	4.997	58.812	349.33	38.442

Track	Mode	Open (deg)	Closed (deg)	Lap (s)	Vmax (km/h)	Wheel work (MJ)
balanced	open-wing	2.000	20.000	59.709	340.10	37.579
balanced	active-ideal	0.000	12.000	59.152	344.32	37.818
balanced	active-limited	0.000	12.000	59.353	349.72	38.667
high downforce	fixed-optimum	18.000	18.000	64.528	334.14	27.316
high downforce	open-wing	2.000	20.000	64.827	337.05	26.513
high downforce	active-ideal	0.000	18.000	64.190	336.19	26.875
high downforce	active-limited	0.000	18.000	64.615	334.89	27.750

RESULT PLATE 12

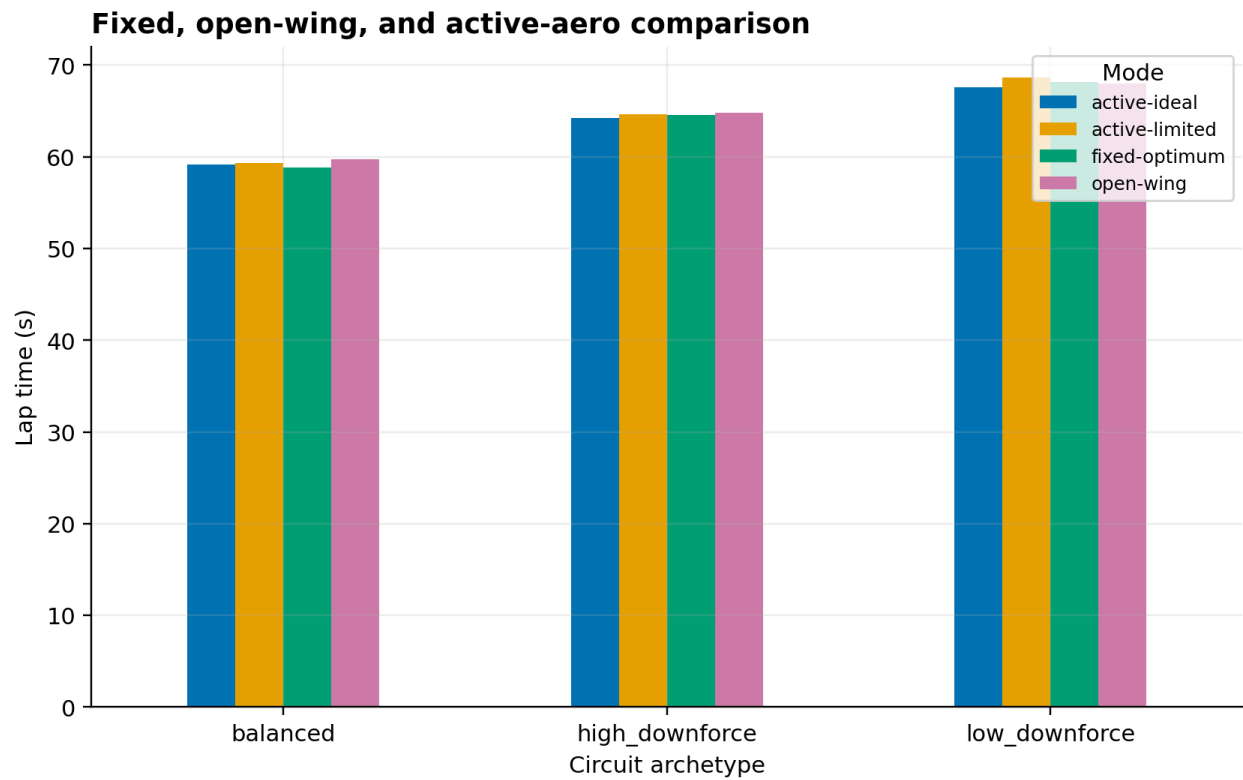


Figure 12. Lap-time comparison of fixed, open-wing, ideal-active, and actuator-limited modes.

Source data: data/strategy_comparison.csv · Units: s · Run: 6bf25b9bebec26e

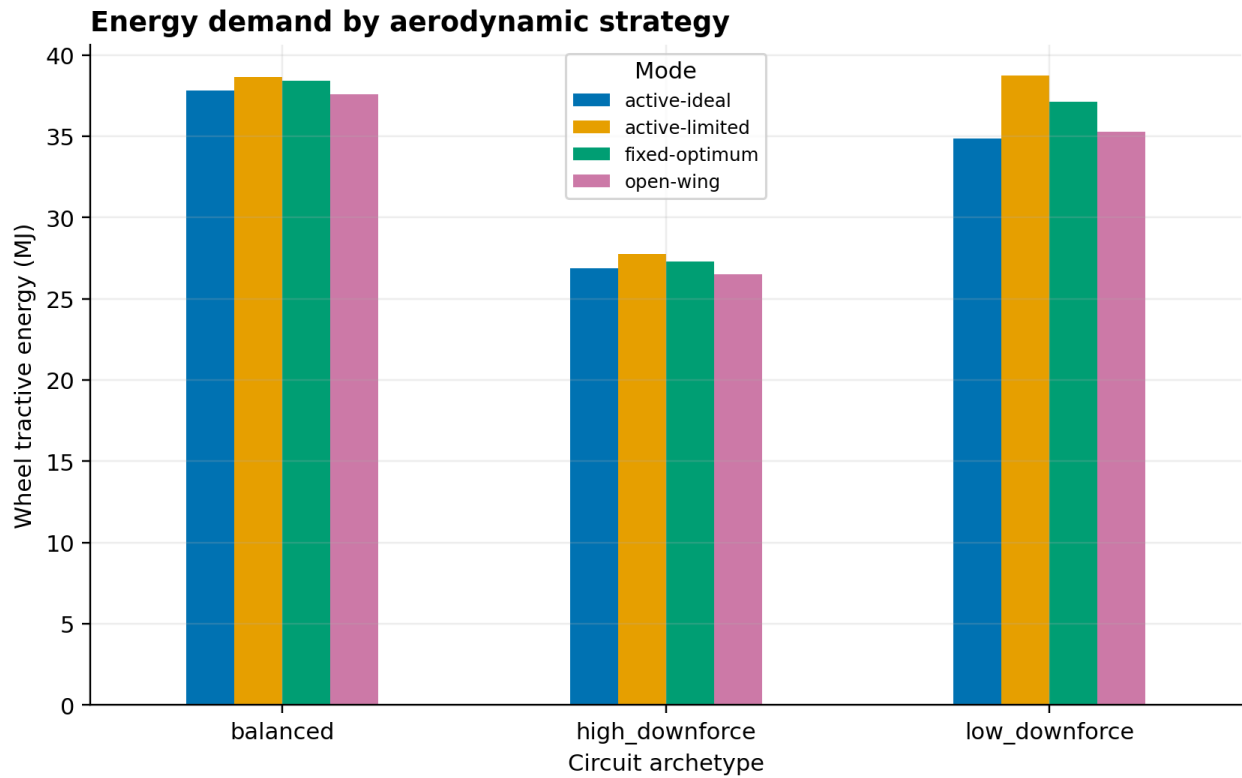
RESULT PLATE 13

Figure 13. Integrated positive wheel work for each aerodynamic strategy.

Source data: data/strategy_comparison.csv · Units: MJ · Run: 6bf25b9bebec26e

7.5 Decision significance

The decision is not solely the mathematical minimum. If the lap-time curve is flat around the optimum, teams can trade a few hundredths of a second for balance, tyre temperature, overtaking speed, or robustness. A steep optimum deserves tighter setup and actuator tolerances. The decision matrix therefore reports gains and penalties by circuit instead of collapsing the project into one universal recommended angle.

RESULT PLATE 16

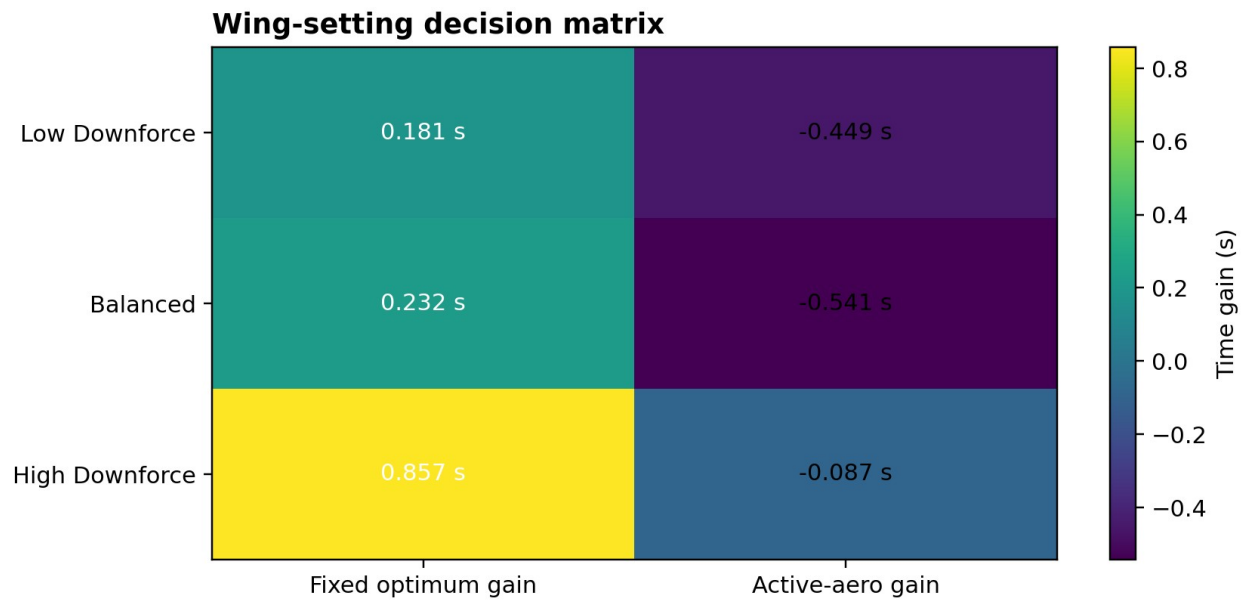


Figure 16. Circuit-by-circuit time gains from fixed-angle optimization and actuator-limited active aero.

Source data: data/decision_matrix.csv · Units: s · Run: 6bf25b9bebec26e

8. Uncertainty and Sensitivity

8.1 Monte Carlo method

One thousand bounded samples per circuit perturb aerodynamic, tyre, mass, powertrain, and environmental parameters. To keep the study computationally practical while retaining full-solver anchoring, a first-order physics response surface is constructed from nominal and perturbed solver cases. The resulting distributions characterize uncertainty in this educational model—not frequencies expected from a real championship car.

8.2 Optimum-angle robustness

The optimum-angle distribution is circuit dependent. Its spread should be read alongside the flatness of the nominal lap-time curve: a wide angle distribution can still correspond to small time loss if multiple settings are nearly equivalent. Conversely, a tight distribution does not prove physical accuracy when all samples share the same simplified model structure.

Table 7. Monte Carlo percentile summary.

Track	Percentile	Metric	Value
low downforce	2.500	optimum angle deg	15.000
low downforce	2.500	minimum lap time s	66.518
low downforce	50.000	optimum angle deg	17.000
low downforce	50.000	minimum lap time s	68.428
low downforce	97.500	optimum angle deg	18.000
low downforce	97.500	minimum lap time s	70.526
balanced	2.500	optimum angle deg	3.000
balanced	2.500	minimum lap time s	57.245
balanced	50.000	optimum angle deg	4.000
balanced	50.000	minimum lap time s	59.008
balanced	97.500	optimum angle deg	5.000
balanced	97.500	minimum lap time s	60.849
high downforce	2.500	optimum angle deg	16.000
high downforce	2.500	minimum lap time s	62.989
high downforce	50.000	optimum angle deg	18.000
high downforce	50.000	minimum lap time s	64.859
high downforce	97.500	optimum angle deg	18.000
high downforce	97.500	minimum lap time s	66.860

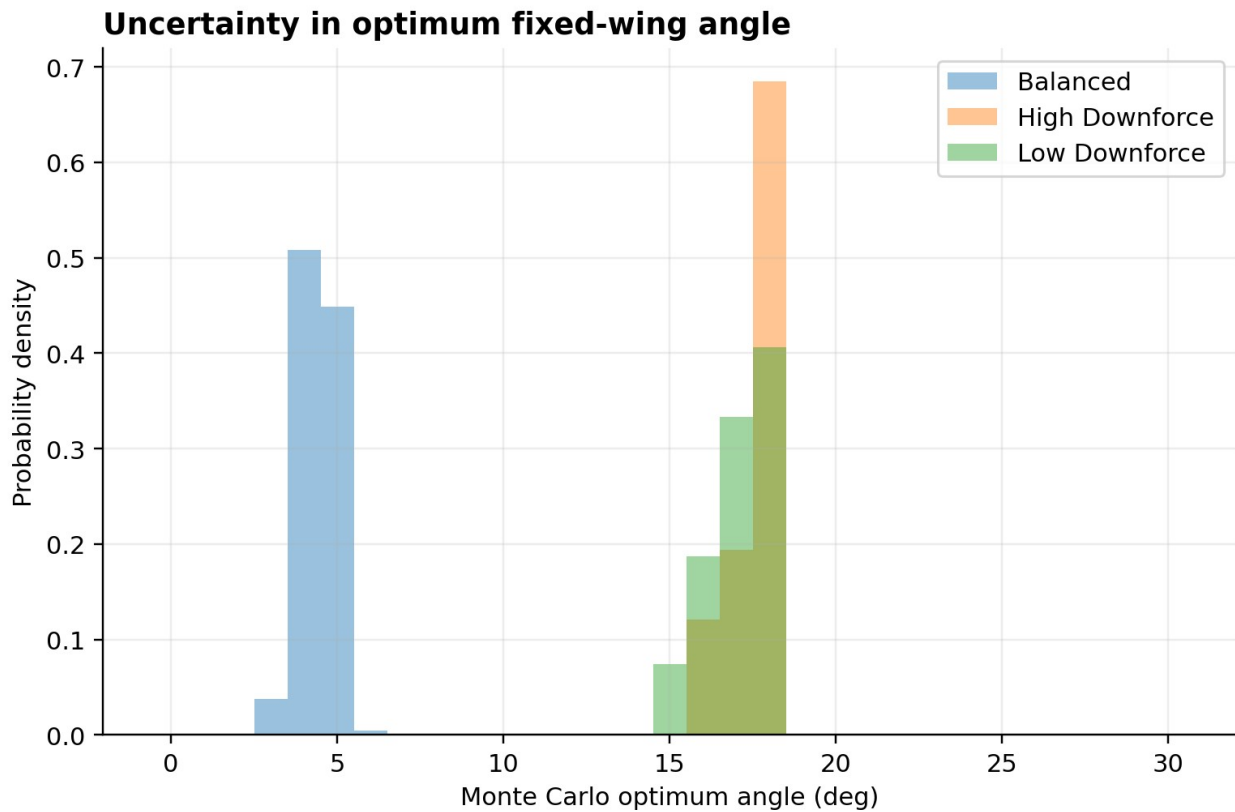
RESULT PLATE 14

Figure 14. Distribution of optimum fixed-wing angle from 1,000 bounded uncertainty samples per circuit.

Source data: data/monte_carlo_samples.csv · Units: probability density · Run: 6bf25b9bebec26e

8.3 Local sensitivity

The balanced-circuit study perturbs selected inputs by plus and minus 10 percent. Air density, lateral tyre friction, and baseline downforce show the largest time changes among the tested terms. The signs require careful interpretation: more density adds both useful downforce and harmful drag; which dominates depends on the circuit and baseline coefficients. The asymmetry between positive and negative changes demonstrates that linear sensitivity is only local.

Table 8. Balanced-circuit local sensitivity.

Parameter	Change (%)	Lap (s)	Delta (s)
environment.air density kgpm3	-10	58.093	-0.739
environment.air density kgpm3	10	60.171	1.339
tyres.mu lateral ref	-10	60.002	1.170
tyres.mu lateral ref	10	58.019	-0.813
tyres.mu longitudinal ref	-10	58.868	0.036
tyres.mu longitudinal ref	10	58.734	-0.098
aero.wing cd0	-10	58.781	-0.051
aero.wing cd0	10	58.883	0.051
aero.baseline cl	-10	59.535	0.703
aero.baseline cl	10	58.446	-0.387

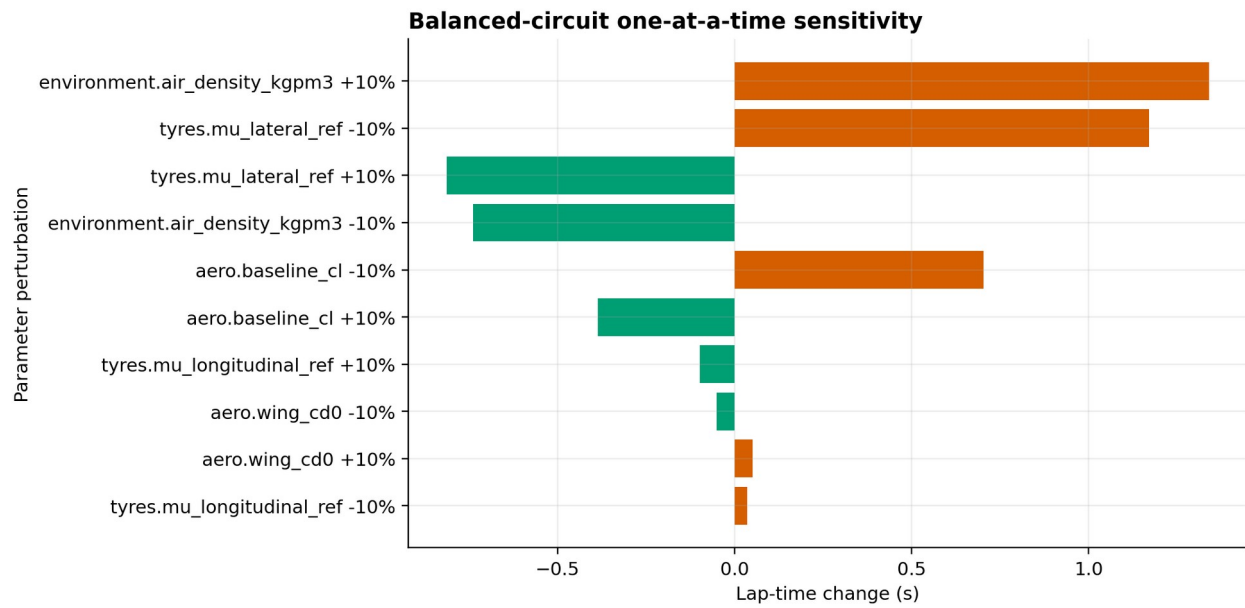
RESULT PLATE 15

Figure 15. Lap-time sensitivity to $\pm 10\%$ input perturbations at the balanced-circuit optimum.

Source data: data/sensitivity.csv · Units: s · Run: 6bf25b9bebec26e

9. Engineering Interpretation

9.1 Why the optimum moves

Rear-wing angle changes several linked quantities at once. More incidence raises rear downforce, changes total aerodynamic balance, increases corner and braking capacity, increases drag, changes achievable gear/speed combinations, and alters energy consumption. A circuit with long straights accumulates drag loss over time, while a circuit with repeated medium/high-speed corners repeatedly harvests the benefit of downforce. Low-speed hairpins depend less on aerodynamics because dynamic pressure is small.

The results also show that track labels cannot substitute for simulation. The synthetic low-downforce circuit still selects a relatively high angle because its particular curvature and braking distribution reward downforce up to the modeled stall boundary. That finding should prompt examination, not manual correction. It demonstrates the value of reporting track definitions and speed traces alongside a headline optimum.

9.2 Setup recommendation logic

For a real engineering program, the computed minimum should define the center of a validation window. Nearby angles would be tested in CFD or a tunnel, then on track, while monitoring front/rear balance, ride height, tyre state, wind, cooling, and actuator behavior. The broad balanced-circuit minimum suggests choosing a robust integer setting around 5 degrees. The 18-degree optima deserve special scrutiny because they coincide with the assumed stall threshold; uncertainty in the actual polar could move them materially.

9.3 Active-aero lesson

The ideal-active benefit and actuator-limited penalty together provide a control-design lesson. The plant has aerodynamic and longitudinal dynamics, but the command has delay and bounded rate. Poor switching can create transient drag without useful cornering load or remove load before braking is complete. Future control should optimize target angles, switching locations, hysteresis, and rate constraints jointly, ideally using predictive control or direct optimal control rather than threshold logic alone.

10. Assumptions and Limitations

The full limitations register is intentionally visible rather than relegated to code comments. Key restrictions are the absence of proprietary geometry and validation data, synthetic circuits, point-mass motion, quasi-steady forces, simplified tyre and powertrain behavior, empirical stall, constant nominal atmosphere, and reduced-order active-aero control. These restrictions mean that absolute lap times and optimum angles must not be presented as values for any real team or venue.

Model-form uncertainty is more important than numerical precision. The solver can repeat a result to many decimal places, but that does not add physical information beyond the assumptions. Reported headline numbers use three decimals for traceability; practical engineering decisions should use wider tolerances. The correct next step is validation, not additional decimal places.

Table 9. Assumptions and limitations register.

ID	Declared limitation
L01	The car is treated as a point mass following a prescribed closed path; yaw dynamics, steering transients, suspension kinematics, heave, pitch, roll, kerbs, traffic, wind, and driver adaptation are outside scope.
L02	Rear-wing angle is the controlled variable. Front-wing, floor, diffuser, cooling, wheel-wake, and body coefficients are collected into fixed baseline coefficients.
L03	Air density is constant within each run. Crosswind and gusts are omitted.
L04	The wing uses a finite-wing lift slope, parabolic induced drag, and a smooth empirical post-stall decay. It is not a CFD- or wind-tunnel-derived multi-element map.
L05	Aero forces vary with dynamic pressure and the declared coefficient model. Ride-height and yaw multipliers exist in the code path but nominal results use zero yaw and nominal ride height.
L06	The tyre model uses load-sensitive friction coefficients and a friction ellipse. It omits temperature, pressure, wear, relaxation length, camber, combined-slip asymmetry, and detailed transient behaviour.
L07	Wheel force is obtained from a representative torque curve, fixed ratios, driveline efficiency, rear-wheel drive, and a redline-limited gear choice. Energy recovery and hybrid deployment are not modeled.
L08	Braking force is constrained by the tyre friction envelope and a global maximum; detailed brake balance, thermal limits, migration, and control-system logic are omitted.
L09	Synthetic track archetypes are used to isolate circuit-character effects. They are not surveyed reproductions of actual Formula 1 venues.
L10	The quasi-steady solver assumes locally equilibrated performance and applies forward/backward spatial passes. It does not integrate a full multibody transient model.
L11	Open and closed wing targets are idealized angle commands within 0–30 degrees.
L12	Ideal-active results are an upper-bound comparison without actuator delay or slew limits.
L13	Actuator-limited results include specified activation/deactivation delays and a 90 degree/s rate limit, but omit motor current, compliance, backlash, sensor filtering, failure modes, and regulatory state logic.
L14	The schedule optimizer is deliberately low dimensional. A slower actuator-limited result than the optimum fixed wing is therefore a valid model outcome, not an error; it indicates that actuation and switching penalties can erase the ideal aerodynamic advantage.
L15	No proprietary geometry, telemetry, CFD field, wind-tunnel data, tyre rig data, or track measurement was available.
L16	MATLAB files are statically checked and compared against

ID	Declared limitation
	Python-exported reference fixtures, but are not runtime-tested in this authoring environment. MATLAB R2022b+ execution is required for final independent parity confirmation.
L17	The Simulink <code>.slx</code> file is generated by <code>build simulink demo.m</code> when a licensed Simulink installation is available; it is a focused longitudinal/aerodynamic demonstration rather than the full spatial lap solver.
L18	The Monte Carlo study uses bounded educational distributions and a first-order response surface anchored to full-solver cases. It characterizes model sensitivity, not real-world probability.
L19	Numerical repeatability is checked through deterministic inputs, a fixed random seed, input/output hashes, contract tests, and cross-format comparisons. Physical validation remains future work.

11. Conclusions and Future Work

11.1 Conclusions

Rear-wing angle produces a nonlinear, circuit-specific performance compromise. Downforce, braking, and cornering benefits grow strongly with speed, while drag reduces acceleration and top speed. The integrated lap model identifies fixed optima of 17.996, 4.997, and 18.000 degrees across the three archetypes. No single setting is universally superior.

The active-aero comparison is equally important. Instantaneous switching can offer a theoretical benefit, but the tested actuator-limited implementation is slower than the best fixed setup. Actuator speed, delay, state selection, and transition timing are therefore first-class performance variables. A movable wing should be assessed as a mechatronic control system, not as two static polars.

The project meets its educational objective by delivering a transparent chain from assumptions to equations, code, tables, figures, uncertainty, MATLAB parity fixtures, and a Simulink builder. Its strongest claim is reproducibility inside the declared model, not real-world prediction.

11.2 Future work

Priority improvements are: replace the analytical polar with full-car CFD or wind-tunnel maps; validate tyre and powertrain data; add pitch, heave, yaw, wind, ride-height coupling, thermal states, and racing-line optimization; model hybrid deployment; perform mesh/time-step and model-form studies; and validate against telemetry. For active aero, introduce constrained optimal control with fail-safe state logic and actuator power/thermal models. MATLAB parity should be executed on R2022b+ and the generated Simulink model should be simulated and archived with logged outputs.

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Appendix A — Parameter Register

The following register distinguishes reference-informed, assumed, and derived inputs. Supported ranges are guards for this educational model, not certification limits.

Table 10. Model parameter register.

Parameter	Unit	Class	Supported range	Description
vehicle.mass kg	kg	reference-informed	740–900	Representative car-plus-driver mass used throughout the comparison.
vehicle.wheelbase m	m	assumed	3.200–3.800	Representative wheelbase for load transfer and aero application geometry.
vehicle.cg height m	m	assumed	0.250–0.400	Centre-of-gravity height used for longitudinal load transfer.
vehicle.front static fraction	fraction	assumed	0.420–0.500	Static normal-load fraction carried by the front axle.
aero.reference area m ²	m ²	assumed	1.200–1.800	Common force-coefficient reference area for the complete car.
aero.wing area m ²	m ²	derived	0.500–0.900	Planform reference area of the simplified rear wing.
aero.stall angle deg	deg	assumed	15–22	Effective angle at the onset of the smooth post-stall model.
aero.baseline cl	dimensionless	assumed	1.500–3.500	Non-rear-wing full-car downforce coefficient magnitude.
aero.baseline cd	dimensionless	assumed	0.600–1.100	Non-rear-wing full-car drag coefficient.
tyres.mu lateral ref	dimensionless	assumed	1.500–2.200	Reference lateral friction coefficient before load sensitivity.
powertrain.torque nm	N m	assumed	450–650	Representative engine torque curve used for comparative simulations.
solver.track spacing m	m	derived	1–5	Maximum spatial step for circuit discretization.

Appendix B — MATLAB and Simulink Workflow

From the project root, add the `matlab` directory to the MATLAB path and run `run_project`. This loads the shared JSON files, sweeps 0–30 degrees, solves all three circuits, and writes MATLAB-native CSV, MAT, and plot outputs. Then run `run_parity_tests` to compare aerodynamic and lap cases with `results/parity/reference_cases.csv`. Relative tolerances are $1e-6$ for coefficients/forces and $1e-4$ for lap metrics.

Run `build_simulink_demo` to generate `F1WingAeroDemo.slx`. The model includes tractive force, a step wing-angle command, a 90 degree/s Rate Limiter, a MATLAB Function aerodynamic block, net-force and inverse-mass blocks, a vehicle-speed integrator, a scope, and workspace logging for speed, angle, downforce, and drag. If Simulink is unavailable the builder exits non-fatally with an instruction. MATLAB source is not runtime-tested in the authoring environment; independent execution requires MATLAB R2022b+ and a Simulink license for the `.slx` model.

Table 11. Software workflow and runtime status.

Layer	Purpose	Status
Python 3.10+	Reference analysis, optimization, uncertainty, exports	Runtime-tested
MATLAB R2022b+	Independent numerical mirror and parity assertions	Static-checked; runtime pending
Simulink	Generated actuator/aero/longitudinal demonstration	Builder supplied; license required
LibreOffice	DOCX/PPTX to PDF conversion	Runtime-tested
pytest	Physics, contract, export, and document checks	Automated

Appendix C — Reproducibility and Artifact Index

Run ID: 6bf25b9bebec26e. Random seed: 3106. The manifest records hashes for every configuration, track, table, figure, workbook, MATLAB data file, and parity fixture. CSV files are the canonical tabular source for figures; the XLSX workbook mirrors them for inspection, while the MAT file supports numerical exchange.

The Python build commands are documented in the README. A clean analysis rebuild regenerates the manifest and numerical assets. The document build consumes that manifest and embeds its run ID and headline values. The final verifier checks file readability, hashes, report and slide structure, and numeric consistency after documented rounding.

Table 12. Manifest artifact index (abbreviated hashes).

Artifact	SHA-256 prefix
data/F1 Wing Angle Results.mat	ac2b01cab933e748...
data/active optimization.csv	8f7cf1d3708d63f1...
data/aero balance.csv	01b29f663784524a...
data/aero polar.csv	157a6d3c87efba06...
data/circuit optima.csv	67fc457726d0c55e...
data/cornering.csv	0b43a9a7c46bf20d...
data/decision matrix.csv	f043ce81377df984...
data/force speed.csv	2f0e3c6dd46f8eb0...
data/lap sweeps.csv	9d8c3ef3884b6e0c...
data/monte carlo correlations.csv	d666fa67ad14bfdd...
data/monte carlo samples.csv	cecf1d8e6484b100...
data/monte carlo summary.csv	4e499890e371f90c...
data/parity cases.csv	c8043f7fe92c90ee...
data/sensitivity.csv	9ee18e29211a2d65...
data/speed traces.csv	c4c3cb77e5489d6c...
data/straight line.csv	648380e236119452...
data/strategy comparison.csv	3b1313f773bca4f2...
figures/01 aero polar.png	5cfe985cc3dd9e49...
figures/01 aero polar.svg	d24266c351c4c7af...
figures/02 aero efficiency.png	caa9f2f5d0a093bb...
figures/02 aero efficiency.svg	606fdbb9e15a12b2...
figures/03 downforce speed.png	c7006eff6dfd523c...
figures/03 downforce speed.svg	94f2daf5f2db3053...
figures/04 drag speed.png	05a648b6df42350f...
figures/04 drag speed.svg	da3aab91cd49c5fd...
figures/05 aero balance.png	dc2d774bb66dc413...
figures/05 aero balance.svg	db2676312b1176f5...
figures/06 top speed.png	4a2485cc32a08588...
figures/06 top speed.svg	bd3632bc72ea1ab5...
figures/07 acceleration.png	7f478396bfd2a582...
figures/07 acceleration.svg	36ab8d225de9bb91...
figures/08 braking.png	01eba75b08d1a698...
figures/08 braking.svg	04688b7ae20e8583...
figures/09 cornering.png	989d8d1ea9e6e418...
figures/09 cornering.svg	fe7007e90ea37036...
figures/10 lap time sweep.png	b9415bd80786f173...
figures/10 lap time sweep.svg	783cf8feaf9d6c13...
figures/11 speed traces.png	60bf7ec899ea9d81...
figures/11 speed traces.svg	16a05d139364af14...
figures/12 strategy comparison.png	b069bf94d846d65a...
figures/12 strategy comparison.svg	20211c0010e2f9f2...
figures/13 energy strategy.png	f389d48bd7eb439c...
figures/13 energy strategy.svg	dd0ab2fda33dae2d...
figures/14 uncertainty optimum.png	d3dda57d85be8e65...
figures/14 uncertainty optimum.svg	a23d55b09a935519...
figures/15 sensitivity.png	ecd6ca4d535101be...
figures/15 sensitivity.svg	e628ee71c0d90305...

Artifact	SHA-256 prefix
figures/16 decision matrix.png	edefaefcb0d5f91c...
figures/16 decision matrix.svg	81ec3aaade42ca6f...
parity/reference cases.csv	c8043f7fe92c90ee...
workbooks/F1 Wing Angle Results.xlsx	d0eed1535e49be44...

Appendix D — Detailed Limitations Statement

This appendix reiterates the interpretation boundary because it is central to responsible use. The car, coefficients, tyre properties, torque curve, track geometry, and actuator behavior are representative assumptions. None is reverse engineered from a named entrant. Plots that resemble engineering deliverables are still reduced-order simulation outputs. They are appropriate for comparing hypotheses and teaching coupled performance analysis; they are not homologation evidence or a substitute for CFD, structural assessment, wind-tunnel work, driver-in-loop evaluation, or track validation.

The project should be cited by model name, version, and run ID. Any modified configuration creates a new evidence set. Conclusions should be restated after every material input change rather than carried forward from the nominal report.